

16.4 dB Outage Coupling Gain Using Branch-Point-Aware Adaptive Optics Over a Deployed Free-Space Optical Link Enabled by Remote Digital Holography

Quinten Moonen, Vincent van Vliet, Menno van den Hout, Eduward Tangdionga, and Chigo Okonkwo
Electro-Optical Communication Group, Department of Electrical Engineering,
Eindhoven University of Technology, the Netherlands, q.h.b.moonen@tue.nl

Abstract We demonstrate for the first time remote digital holographic wavefront sensing over a deployed free-space optical link during a 90-hour measurement campaign. Complex-field recovery enables turbulence estimation and branch-point-aware digital adaptive optics, providing 16.4-dB outage gain in our system over branch-point-unaware reconstruction under strong turbulence. © 2026 The Author(s)

Introduction

Free-space optical (FSO) communication is considered a key technology for future high-capacity satellite downlinks, offering orders-of-magnitude greater throughput than radio-frequency systems due to increased bandwidth and directivity. However, FSO downlinks suffer from atmospheric effects such as optical absorption, scattering, and turbulence [1, 2]. Optical turbulence disturbs the received wavefront, causing scintillation and severe fibre-coupling loss at the receiver, reducing link availability. Adaptive optics (AO) is the primary approach for mitigating atmospheric-turbulence-induced wavefront distortions and restoring fibre-coupling efficiency in FSO receivers to increase link performance and availability. In a closed-loop AO system as shown in Fig. 1, a wavefront sensor (WFS) estimates the aberration of the incoming field and a corrective element such as a deformable mirror (DM) compensates for the distortions. Consequently, the performance of the AO system is fundamentally determined by the accuracy and completeness of the wavefront measurement. Most AO receivers rely on a Shack-Hartmann WFS (SH-WFS) [1, 2], which reconstructs the phase from local wavefront slopes. While effective under moderate turbulence, such slope-based reconstructors fail in the presence of branch points and the corresponding hidden phase, arising from phase singularities [3–5] associated with strong turbulence conditions. As a result, significant residual aberrations remain uncompensated, leading to reduced fibre-coupling efficiency and degraded link performance. Direct measurement of the optical complex field offers a route to overcoming this limitation. Interferometric WFSs, such as the digital holography WFS (DH-WFS), give access to this full complex field, enabling explicit reconstruction of the branch points and hidden phase structures that are inaccessible to conventional slope-based WFSs, and have been shown to improve AO performance [4, 5]. Despite its advantages, the conventional DH-WFS requires a coherent reference derived from the

transmitter, making practical deployment in satellite downlinks challenging. Recently, remote digital holography has emerged as a promising alternative by regenerating the reference locally from the received light through optical injection locking (OIL) [6, 7], and in [8] we demonstrated the compatibility with FSO.

In this work, we experimentally demonstrate remote digital holographic wavefront sensing in a deployed terrestrial FSO communication testbed operated over three days, as an accessible proxy for the satellite downlink. Using the recovered complex optical field, we estimate channel turbulence parameters and validate the results against scintillometer measurements. We further implement a digital branch-point-aware AO receiver and evaluate both mean and outage fibre-coupling performance. The measurements reveal a strong dependence of AO gain on turbulence strength and show that branch-point-aware correction substantially improves the tail of the coupling distribution. Compared with conventional branch-point-unaware sensing, the proposed approach achieves up to 16.4 dB of coupling gain at an outage probability of 10^{-3} , highlighting its potential to mitigate deep-fade events in future FSO satellite downlinks.

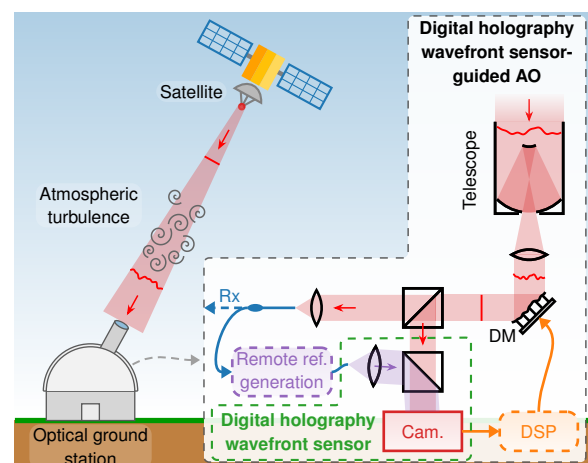


Fig. 1: Schematic diagram of a DH-WFS-guided AO system using remote reference generation for turbulence mitigation in an FSO satellite downlink.

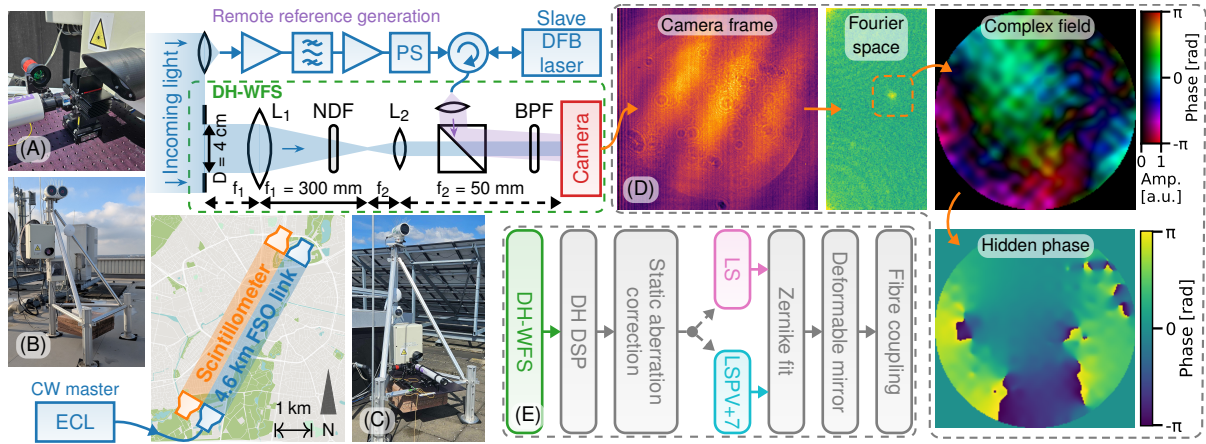


Fig. 2: Experimental setup using the remote DH-WFS in a deployed FSO link (A) DH-WFS (B) Tx optical terminal (C) Rx optical terminal and the DH-WFS (D) DSP steps to extract the complex field and hidden phase from the camera frame (E) AO pipeline

Remote Digital Holography

Digital holography provides direct access to the complex optical field through a single-shot interference measurement between a signal and a reference wave [9]. As illustrated in Fig. 2d, the recorded off-axis interferogram is processed in the Fourier domain, where the signal-reference sideband is isolated to reconstruct the amplitude and phase of the received wavefront. Unlike slope-based wavefront sensing, this approach directly captures hidden phase structures and branch points. A key challenge for applying digital holography to satellite downlinks is the generation of a phase-coherent reference at the receiver. To address this, remote digital holography employs optical injection locking (OIL) to regenerate the reference directly from the received communication signal [8]. The received light is amplified, filtered, and injected into a slave laser, which, once locked, preserves the phase information of the received field while providing sufficient optical power for holographic detection. The regenerated field is subsequently used as the reference beam in the DH-WFS, enabling coherent complex-field sensing without distributing an optical reference from the transmitter. This capability removes a major obstacle to deploying holographic wavefront sensing in adaptive-optics-assisted FSO satellite links.

Remote Digital Holography Wavefront Sensor in the Terrestrial FSO Testbed

The DH-WFS was installed in a 4.6 km terrestrial FSO testbed [10], shown in Figs. 2a to 2c. A narrow-linewidth laser (<100 kHz) was amplified to 30 dBm and transmitted through the FSO channel. Part of the received signal was coupled into a thermally-expanded-core single-mode fibre (SMF) and used to generate a local reference through optical injection locking. This enables holographic wavefront sensing without distributing a reference from the transmitter, a key requirement for future satellite links.

Installed adjacent to the receiving terminal as shown in Fig. 2c, the DH-WFS captures a fraction

of the received beam from the FSO link. A Keplerian beam reducer with a 4 cm diameter collecting aperture demagnifies the captured wavefront in the pupil plane onto the camera. To avoid overexposure and block sunlight, a neutral-density filter attenuates the signal and a 12 nm bandpass filter is placed before the camera. A beamsplitter combines the reference beam with the demagnified signal in an off-axis holographic configuration before reaching the camera sensor. Frames were acquired at 334 fps with a 25 μ s exposure time. Throughout the multi-day campaign, periodic short and long captures were recorded to continuously monitor atmospheric conditions.

In the subsequent DH digital signal processing (DSP), shown in Fig. 2d, the pupil is detected in the camera frame, cropped, and zero-padded. A 2D fast Fourier transform (FFT) is applied to separate and filter the signal-reference product. The signal-signal and reference-reference products at DC are filtered away, the sideband is located, and the complex wavefront is reconstructed via an inverse fast Fourier transform (IFFT). To isolate turbulence-induced distortions from setup perturbations, the ensemble-average phasor of each capture is removed, assuming zero-mean turbulence.

The turbulence parameters of each capture are obtained by taking the 2D mutual coherence function (MCF) of the complex fields using an FFT-based autocorrelation approach [11]. Per frame, the radial average of the normalised MCF is taken, and subsequently, the capture ensemble average is taken. This is fitted to a Kolmogorov power law to extract the Fried parameter r_0 . In parallel with the FSO link, a scintillometer system is deployed to independently measure optical turbulence.

To evaluate the benefits of full complex-field sensing, digital AO was applied using a digital DM model with Zernike modes $J \leq 21$ as the wavefront corrector. As continuous surface DMs cannot physically apply the conjugate of the wrapped-phase discontinuities, the phase is first unwrapped. To compare branch-point-unaware and -aware correc-

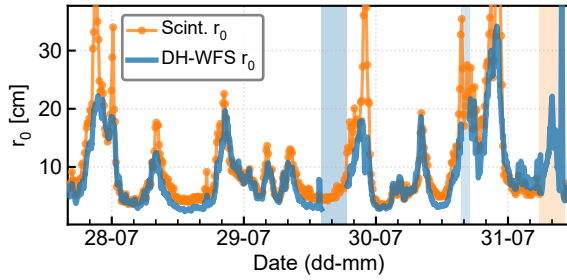


Fig. 3: Measured Fried parameter using the DH-WFS and the scintillometer over the 90-hour measurement.

tion, a least-squares (LS) unwrapped phase reconstructor, blind to the branch-point phase, was used as the SH-WFS-equivalent, while post-processing congruence operation (PCO) recovers the true field by adding the wrapped hidden phase to the LS phase. This is further improved through the LS principal value (LSPV+7) algorithm [12]. By applying an optimised piston to the field, the branch cuts are placed in areas of low intensity to minimise the DM fitting error penalty. The corrected fields were finally evaluated based on their coupling efficiency into the LP₀₁ mode of a SMF.

Turbulence Estimation Results

The remote DH-WFS operated continuously for more than 90 hours, enabling long-term characterisation of the atmospheric channel. The estimated turbulence evolution is shown in Fig. 3, where the shaded regions correspond to interruptions caused by scheduled testbed maintenance and periods of heavy rainfall. Apart from these events, the sensor provided stable operation throughout the measurement campaign.

The Fried parameter r_0 estimates from the DH-WFS show good agreement with the scintillometer across a wide range of atmospheric conditions. The median signed relative differences in r_0 between the scintillometer and the DH-WFS are -22.2% , -19.3% , and -11.7% for weak ($D/r_0 < 0.5$), moderate ($0.5 \leq D/r_0 < 1$), and strong ($D/r_0 \geq 1$) turbulence regimes, respectively. Here, D denotes the receiver aperture diameter. Notably, the discrepancy decreases with increasing turbulence strength, indicating improved convergence in the regime most relevant for AO-assisted FSO links. The systematically lower r_0 estimates obtained by the DH-WFS can be attributed to differences in both measurement principle and path weighting. For weak turbulence, the relatively small aperture limits the observable decay of spatial coherence, causing the Kolmogorov fit to underestimate large coherence lengths. Furthermore, whereas the scintillometer reports a path-weighted estimate of C_n^2 with peak sensitivity near the link midpoint, the DH-WFS directly measures the coherence properties of the received field and is therefore more sensitive to turbulence close to the receiver under spherical-wave propagation.

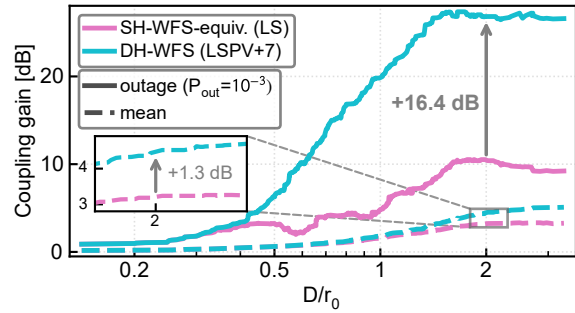


Fig. 4: Mean and outage AO coupling gain using different phase compensation algorithms vs. the turbulence strength.

Adaptive Optics Gain Results

The fibre-coupling performance of the different phase-compensation architectures is summarised in Fig. 4, where the coupling gain is defined by the loss difference between the compensated and uncompensated wavefronts. For strong turbulence conditions, branch-point-aware correction using the DH-WFS-guided LSPV+7 algorithm provides a mean coupling gain of 1.3 dB relative to conventional branch-point-unaware LS reconstruction. While the average gain is modest, the primary performance metric for practical FSO systems is outage behaviour rather than mean coupling efficiency. To quantify this effect, we evaluate the outage coupling loss, defined as the loss exceeded with probability 10^{-3} , corresponding to the 99.9th percentile of the per-capture loss distribution. In this regime, the benefit of full complex-field sensing becomes substantially more pronounced as these deep fades carry more phase distortions and hidden phase, precisely when slope-based WFSs fail. As shown in Fig. 4, branch-point-aware correction with the remote DH-WFS improves outage coupling by up to 16.4 dB over gradient-based (LS) sensing, showing that access to the full complex field is key to mitigating the rare, severe fades that dominate link availability in turbulence-limited FSO communication links.

Conclusions

We demonstrated the first remote digital holographic wavefront sensor in a deployed free-space optical communication link, with measurements spanning more than three days of varying atmospheric conditions. Accurate turbulence estimation was obtained directly from the recovered complex optical field and validated against scintillometer data. Through digitally emulated adaptive optics, branch-point-aware correction yielded up to 16.4 dB outage coupling gain at an outage probability of 10^{-3} over conventional wavefront sensing under strong turbulence in our system. Our results demonstrate a practical pathway for deploying full complex-field wavefront sensing in future adaptive-optics-assisted FSO satellite downlinks, improving their resilience against deep turbulence-induced fading and thus link availability.

Acknowledgements

This work was supported in part by the Dutch Research Council (NWO) TTW-Perspectief Project Optical Wireless Superhighways: Free photons (FREE) under Grant P19-13, in part by the Dutch Ministry of Economic Affairs and Climate Policy (EZK) through the PhotonDelta National Growth Fund (NGF) Programme on Photonics, in part by the European Innovation Council Transition project CombTools under Grant G.A. 101136978, in part by the Quantum Delta NL CAT-2: National Quantum Network program, and in part by the NWO NGF project STARS: Smart 3D Tomographic Adaptive optics for Turbulence Resilient optical communications System. We thank Aircision B.V., particularly Nourdin Kaai, Luis Pellicer Collado, Juan Andrés Apolo, and Roland Blok, for their support of the Reid Photonloop testbed.

References

- [1] B. I. Bitachon et al., 'Tbit/s single channel 53 km free-space optical transmission-assessing the feasibility of optical geo-satellite feeder links', in *European Conference and Exhibition on Optical Communication*, Optica Publishing Group, 2022, Th3A–6.
- [2] R. Boddeda et al., 'Proof of concept of 100 gigabit/s per carrier optical transmission from the moon with a turbulent channel replicator and adaptive optics', in *IET Conference Proceedings CP839*, IET, vol. 2023, 2023, pp. 1694–1697. DOI: 10.1049/icp.2023.2669.
- [3] D. L. Fried, 'Branch point problem in adaptive optics', *J. Opt. Soc. Am. A*, vol. 15, no. 10, pp. 2759–2768, Oct. 1998. DOI: 10.1364/JOSAA.15.002759. [Online]. Available: <https://opg.optica.org/josaa/abstract.cfm?URI=josaa-15-10-2759>.
- [4] M. Kalensky, M. T. Banet, T. J. Bukowski, E. M. Bates, M. W. Hyde IV, and M. F. Spencer, 'Benchtop implementation of branch-point-tolerant adaptive optics', *Applied Optics*, vol. 65, no. 19, H1–H13, 2026. DOI: 10.1364/AO.581692.
- [5] M. T. Banet and M. F. Spencer, 'Compensated-beacon adaptive optics using least-squares phase reconstruction', *Optics express*, vol. 28, no. 24, pp. 36 902–36 914, 2020.
- [6] A. Kawai, S. Shimizu, K. Shibahara, T. Kobayashi, and Y. Miyamoto, 'Remote Digital Holographic Characterization of a 75.2 km Few-Mode Fiber without Reference Wave Transfer', in *2023 Optical Fiber Communications Conference and Exhibition (OFC)*, 2023, M4B.4. DOI: 10.1364/OFC.2023.M4B.4.
- [7] D. Shaji et al., 'Characterization of mode-dependent gain in distributed-raman-amplified 90-km multi-mode fiber with remote digital holography', in *European Conference and Exhibition on Optical Communication*, Optica Publishing Group, 2026, We2–H3.
- [8] Q. Moonen, M. van den Hout, V. van Vliet, S. V. Andreu, and C. Okonkwo, 'Complex field characterization for free-space optical communications using remote digital holography', in *CLEO 2026*, Optica Publishing Group, 2026, STH1J.3. DOI: 10.1364/CLEO_SI.2026.STH1J.3.
- [9] S. van der Heide et al., 'Optical Field Characterization using Off-axis Digital Holography', in *Optical Fiber Communication Conference (OFC) 2022*, Optica Publishing Group, 2022, M3Z.6. DOI: 10.1364/OFC.2022.M3Z.6.
- [10] V. van Vliet, M. van den Hout, K. Gümüş, E. Tangdionga, and C. Okonkwo, 'Multi-Terabit Coherent Free-Space Optical Communication Over a 4.6 km Urban Channel', *Journal of Lightwave Technology*, 2026. DOI: 10.1109/JLT.2026.3676755.
- [11] J. Schmidt, *Numerical Simulation of Optical Wave Propagation with Examples in MATLAB* (Press monograph). SPIE, 2010, ch. 3, 9, ISBN: 9780819483263. DOI: 10.1117/3.866274. [Online]. Available: <https://books.google.nl/books?id=RqPZSAAACAAJ>.
- [12] M. J. Steinbock, M. W. Hyde, and J. D. Schmidt, 'Lspv+7, a branch-point-tolerant reconstructor for strong turbulence adaptive optics', *Applied Optics*, vol. 53, no. 18, pp. 3821–3831, 2014. DOI: 10.1364/AO.53.003821.